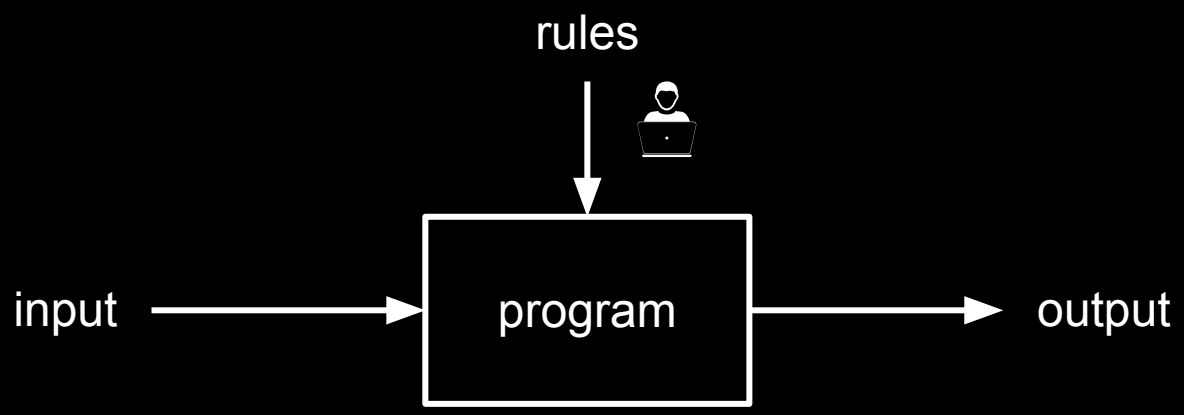
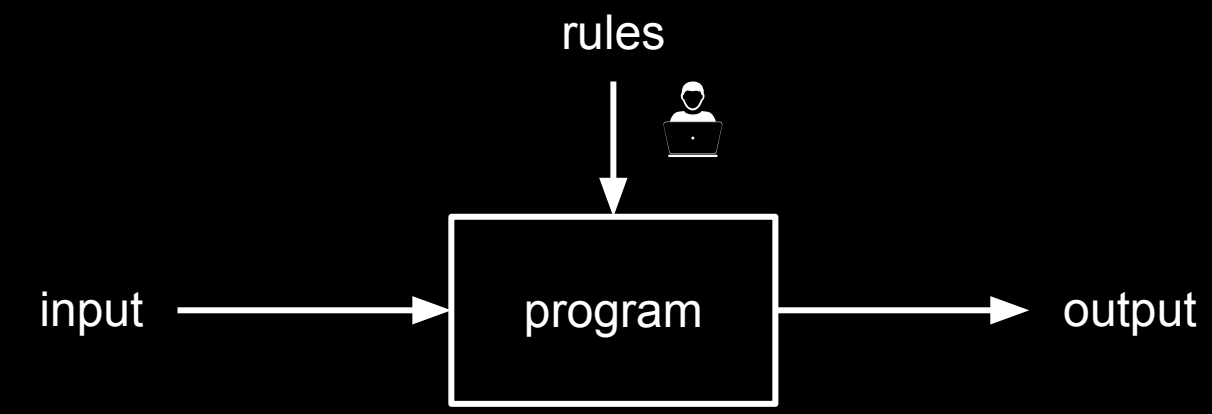
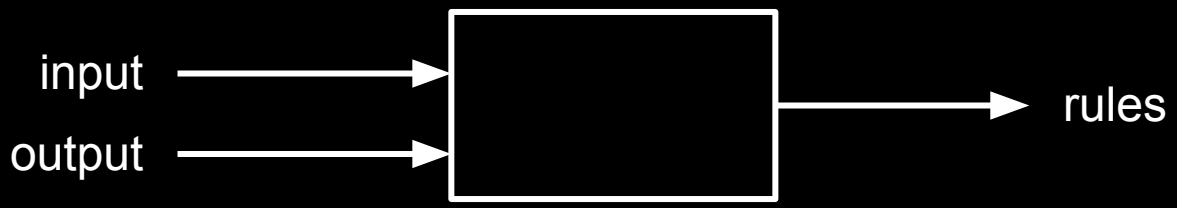
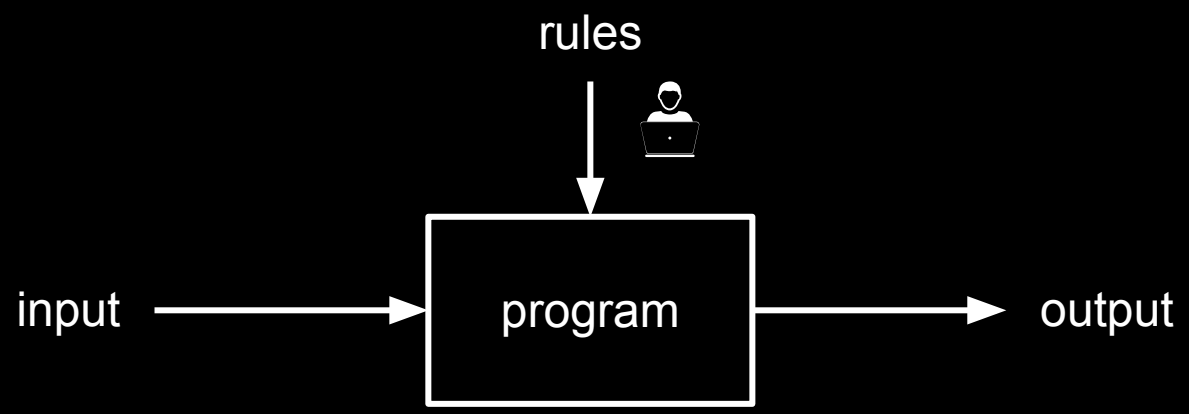
LEARNING

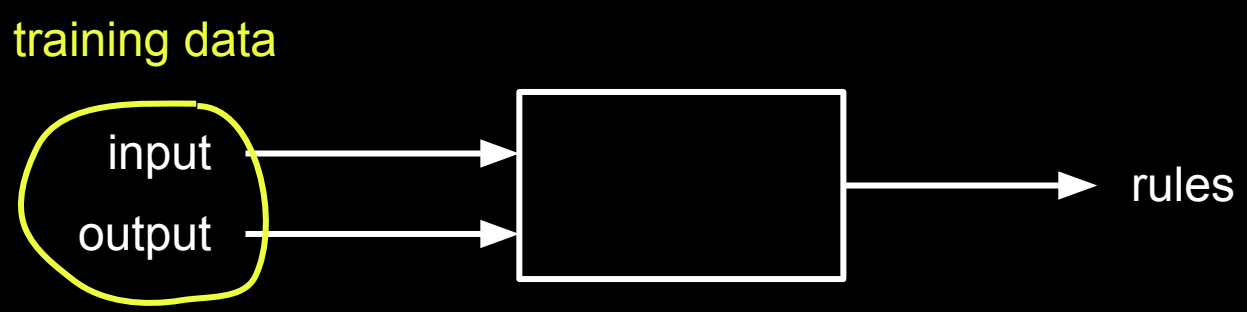
input
program
output

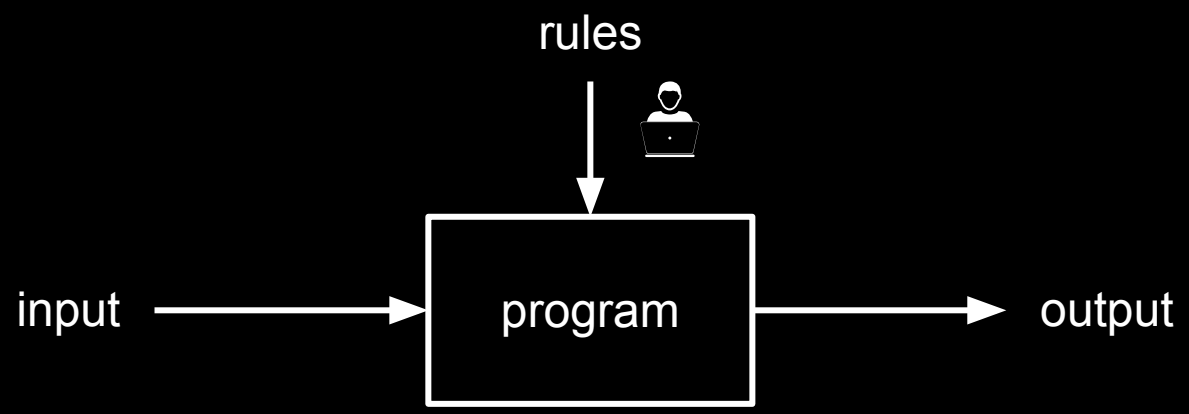

rules
input
program
output


rules
input
program
output

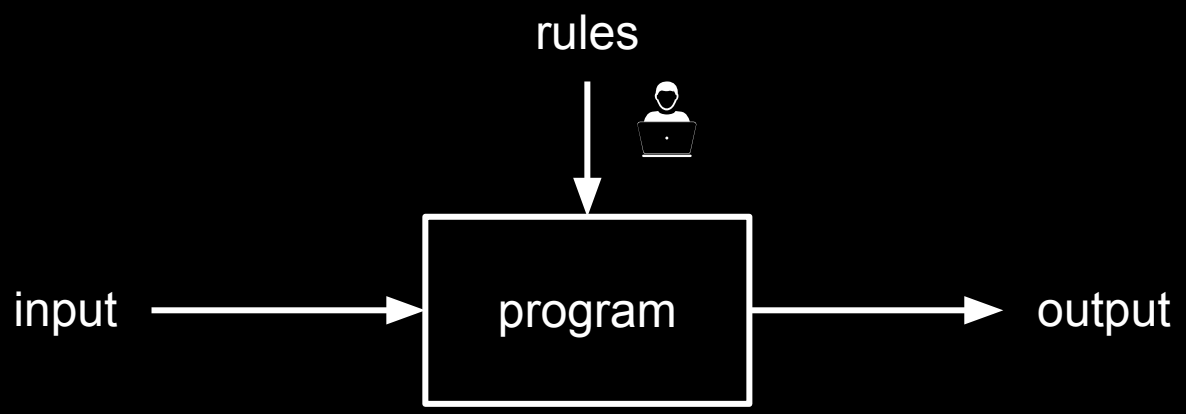

input
output
rules

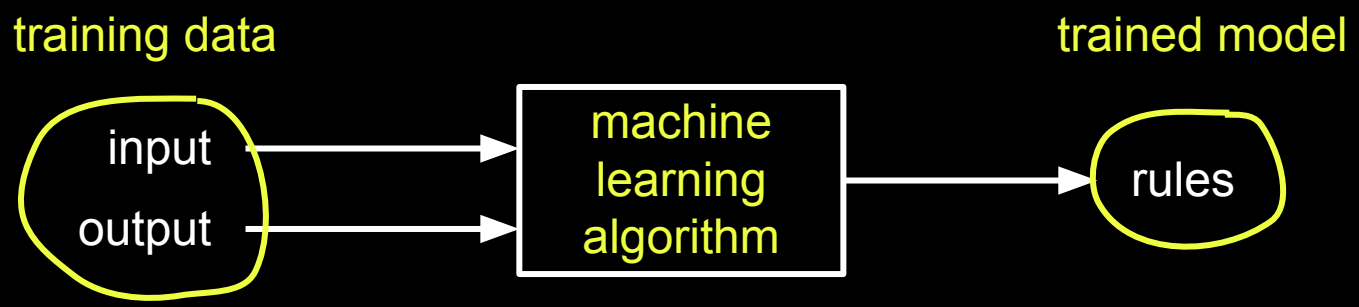

rules
input
program
output

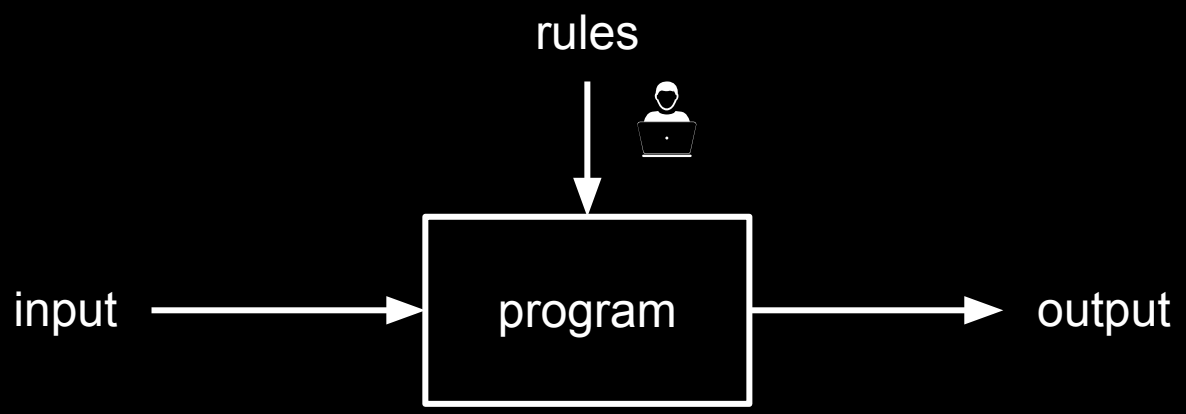

training data
input
output
rules

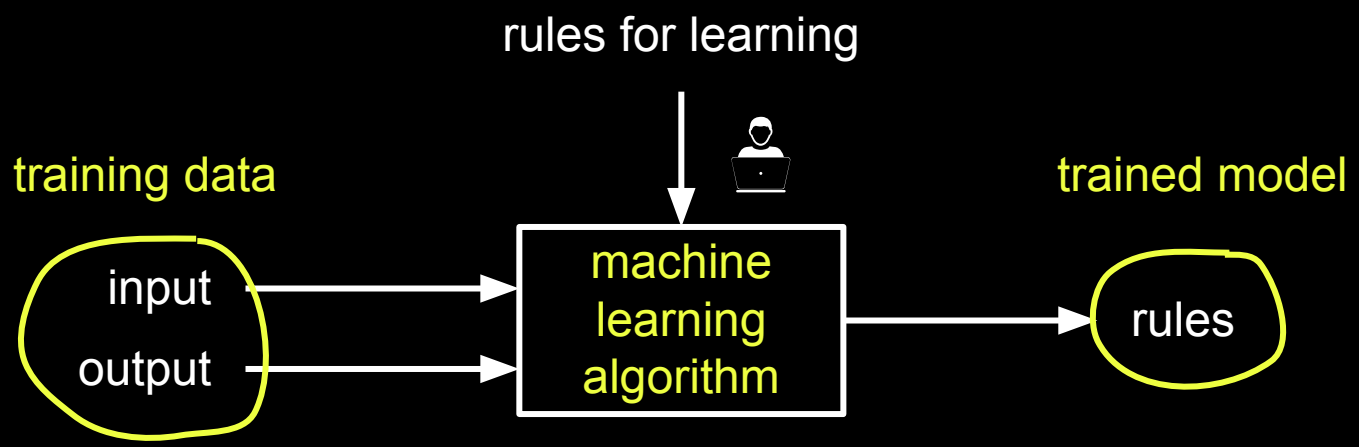

rules
input
program
output

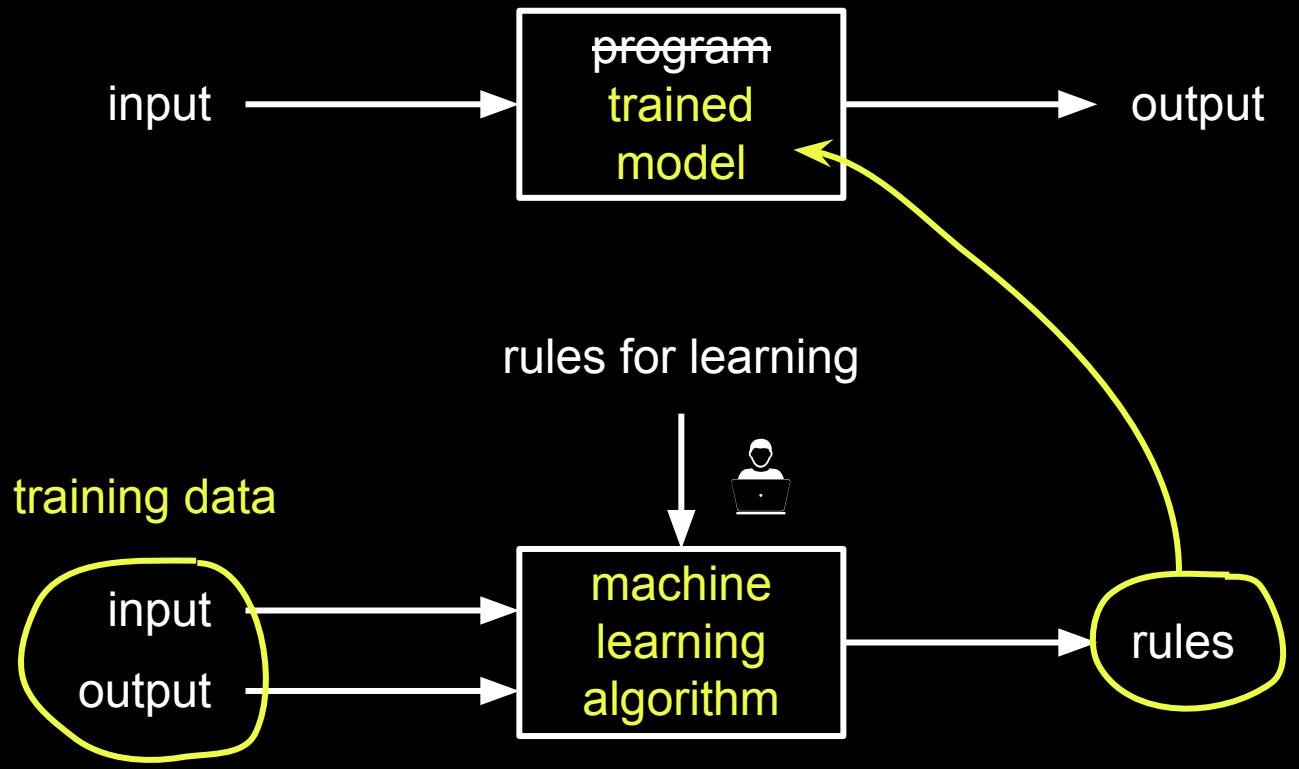
training data
input
output
trained model
rules

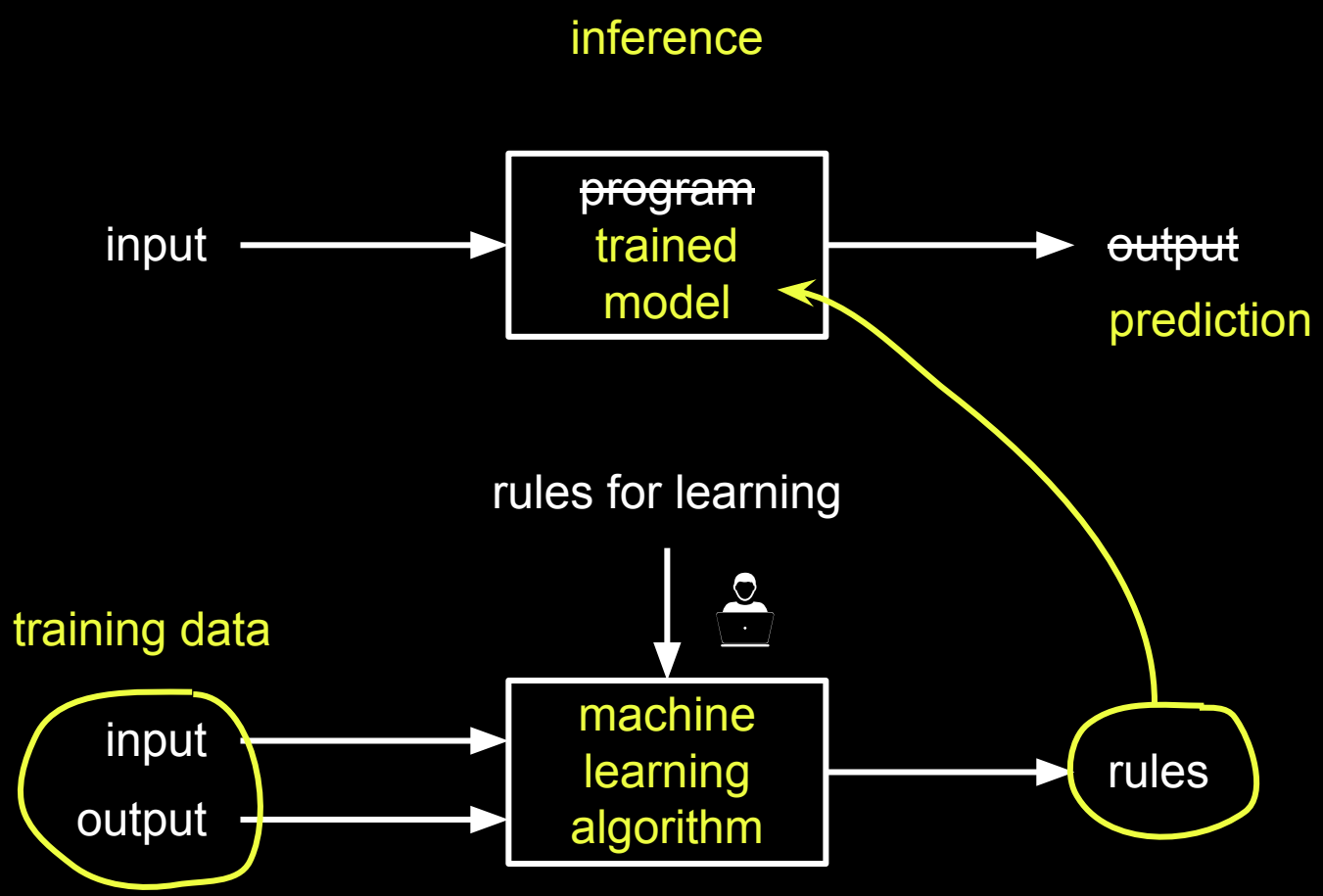

rules
input
program
output

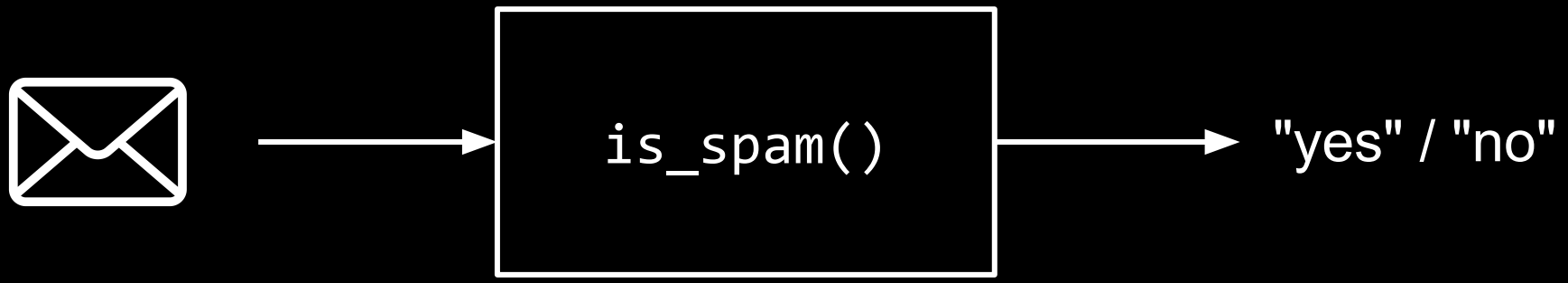

training data
input
output
machine learning algorithm
trained model
rules

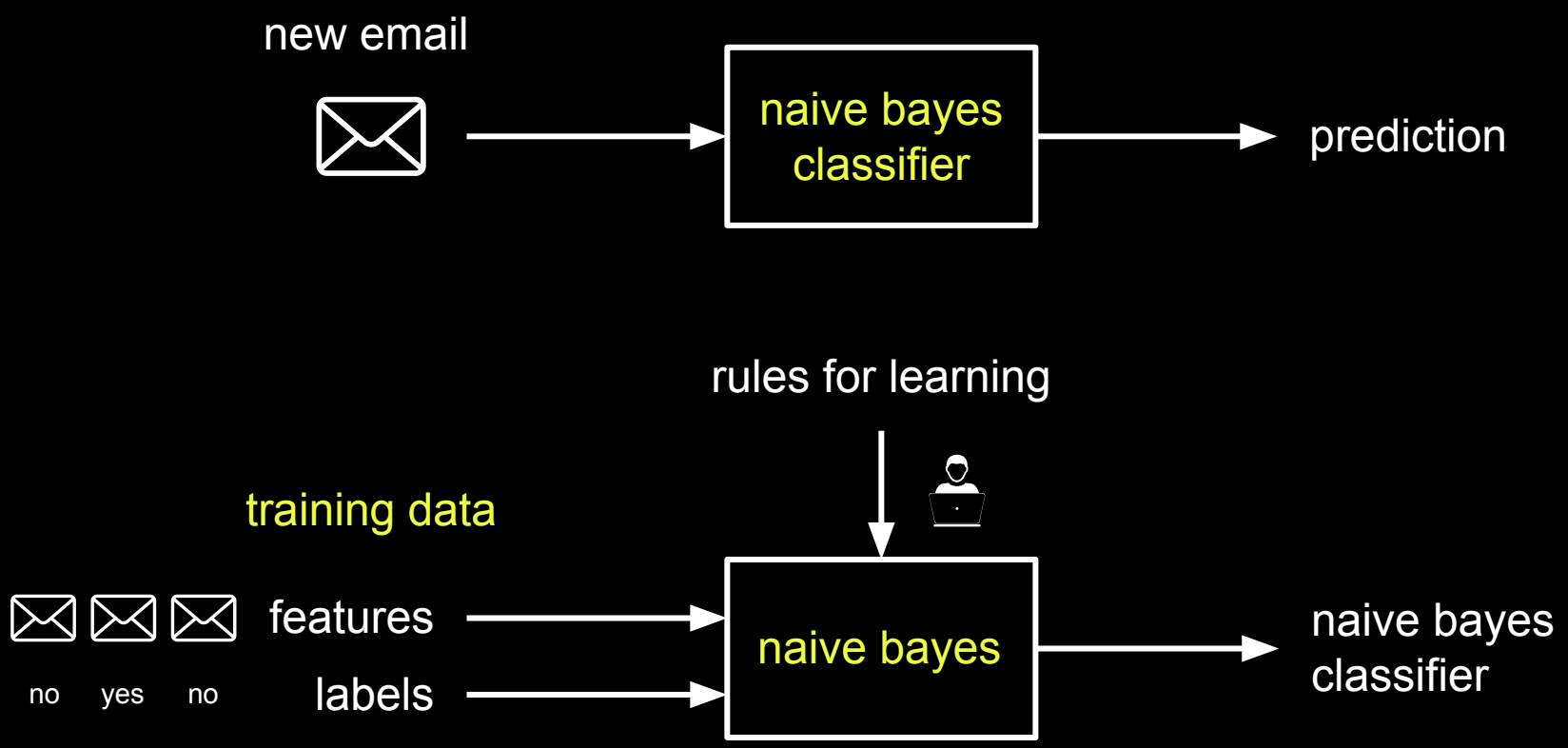

rules
input
program
output


rules for learning
training data
input
output
machine learning algorithm
trained model
rules

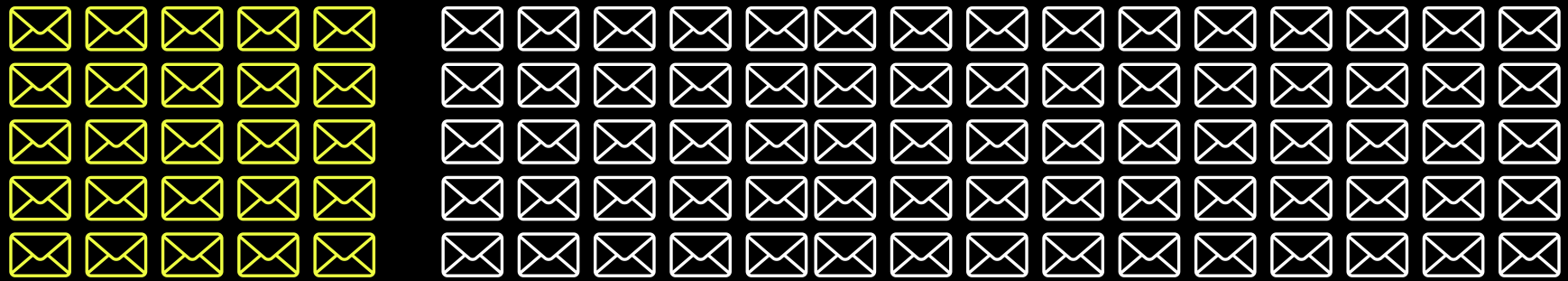

input
~~program~~
trained
model
output
rules for learning
training data
input
output
machine
learning
algorithm
rules


inference
input
~~program~~
trained
model
~~output~~
prediction
rules for learning
training data
input
output
machine
learning
algorithm
rules


is_spam()
"yes" / "no"


new email
naive bayes classifier
prediction
rules for learning
training data
no
yes
no
features
labels
naive bayes
naive bayes classifier

$$P(spam|words) = \frac{P(words|spam) \times P(spam)}{P(words)}$$



	spam		no spam	
total	25		75	
"buy"	20	20/25 = 0.8	5	5/75 ≈ 0.066
"cheap"	15	15/25 = 0.6	10	10/75 ≈ 0.133
"buy" & "cheap"	12	0.8 × 0.6 = 0.48	≈ 0.67	0.066 × 0.133 ≈ 0,0089

P(spam | buy & cheap) = 12 / (12 + 0.67) ≈ 0.947

I. we can write rule-based programs to solve some problems

II. some problems are too complex to be solved with rules alone

III. for some problems, we cannot define a suitable set of rules to solve it

IV. for some problems, we can have a computer program learn the rules from data (machine learning)

x_1	x_2	y
4	2	8
1	2	5
0	5	10
2	1	4
3	2	?

Source: Ananthaswamy, Anil. Why Machines Learn: The Elegant Math behind Modern AI. Dutton, 2024.

x_1	x_2	y
4	2	8
1	2	5
0	5	10
2	1	4
3	2	?

Source: Ananthaswamy, Anil. Why Machines Learn: The Elegant Math behind Modern AI. Dutton, 2024.

which function produces the observed pattern?

$$f(x_1, x_2)$$

Source: Ananthaswamy, Anil. Why Machines Learn: The Elegant Math behind Modern AI. Dutton, 2024.

in this case, it's easy to spot
a potential candidate

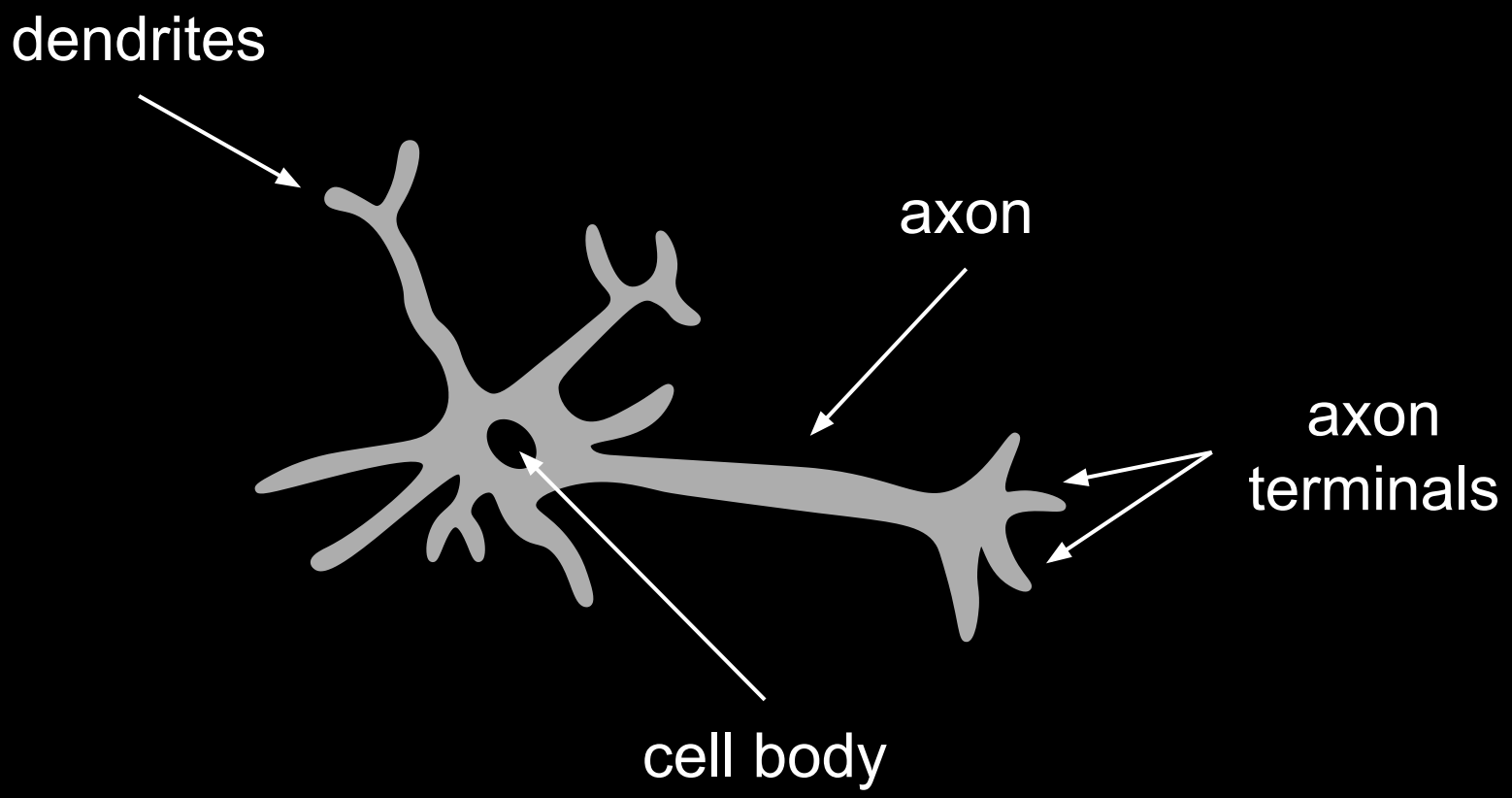
$$y = x_1 + 2x_2$$

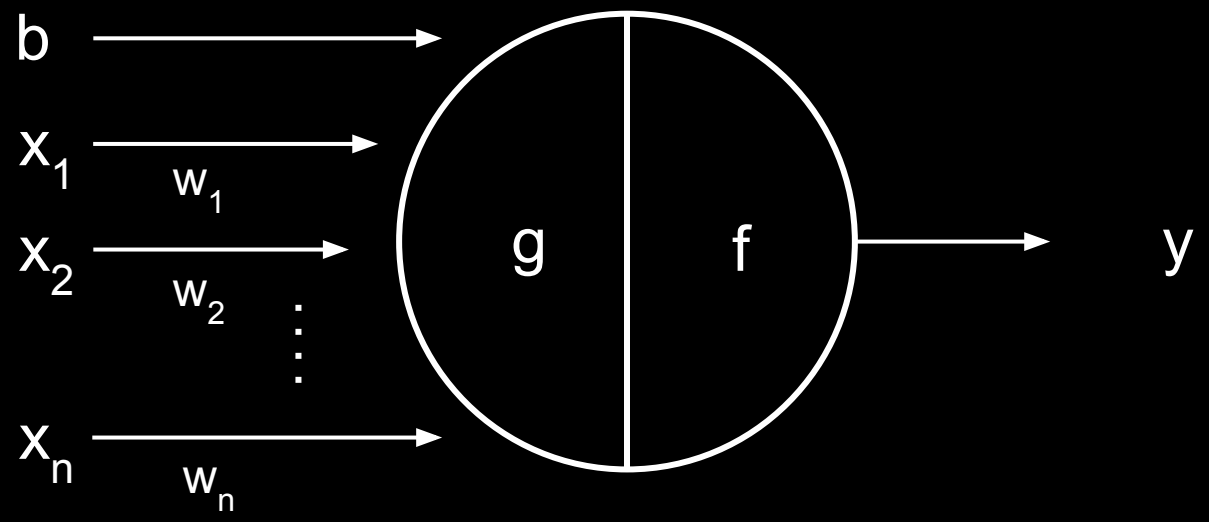
Source: Ananthaswamy, Anil. Why Machines Learn: The Elegant Math behind Modern AI. Dutton, 2024.

$$y = w_1 x_1 + w_2 x_2$$

what if it's not so easy?

Image generated with Flux Pro by Black Forest Labs


dendrites
axon
axon
terminals
cell body



perceptron

multi-layered
perceptron

backpropagation

is my password safe?